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Data migration synchronization process using a manifest from a source device to a new destination device

Abstract

In some implementations, a computing device may configure a new device based on a current state of an old device, including settings, preferences, and other user data. The data may be transferred from the old device to the new device, and then relocated according to a manifest that details positions of the data on the old device. The destination device may be rebooted into a configuration mode to allow for the relocation of the transferred data, and then rebooted again to configure the destination device to provide access to the data in its respective relative locations on the destination device.

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Background/Summary

INCORPORATION BY REFERENCE; DISCLAIMER (1) Each of the following applications are hereby incorporated by reference: application Ser. No. 18/338,308 filed on Jun. 20, 2023;

application Ser. No. 17/452,850 filed on Oct. 29, 2021; application Ser. No. 16/677,530 filed on Nov. 7, 2019; application No. 62/854,880 filed on May 30, 2019. The Applicant hereby rescinds any disclaimer of claim scope in the parent application(s) or the prosecution history thereof and advises the USPTO that the claims in this application may be broader than any claim in the parent application(s).

TECHNICAL FIELD

(1) The disclosure generally relates to initial configuration of an electronic device, and more particularly, to automatically migrating data from an old device to a new device.

BACKGROUND

(2) When a person purchases a new electronic device, the person typically desires to have all of the data from his or her old device be transferred to the new device. Utilizing a third-party service to transfer the data between the devices may lead to privacy and content ownership issues and also may be burdensome on the person should the person need to monitor the progress of the transfer and provide input to continue the process at certain points. Moreover, the data transfer via the third-party service may not be performed in a secure, efficient, and optimal manner. Also, sometimes a data network connection may not be available for performing the transfer, or may not be desired, due to privacy concerns. Furthermore, some data may not be transferred and lost in the process, which may be view as a serious breach of the person's trust in the manufacturer of the electronic device. However, most conventional configuration routines for new electronic devices do not address all of these concerns.

SUMMARY

(3) In some implementations, a computing device may configure a new device based on a current state of an old device, including settings, preferences, and other user data. The data may be transferred from the old device to the new device, and then relocated according to a manifest that details positions of the data on the old device. The destination device may be rebooted into a configuration mode to allow for the relocation of the transferred data, and then rebooted again to configure the destination device to provide access to the data in its respective relative locations on the destination device.

(4) Particular implementations provide at least the following advantages. The new device may be set up automatically in accordance with the configuration of the old device, thereby allowing a user to use the new device immediately upon acquiring the new device, without needing to re-install applications used by the user on the old device and reconfigure all the settings and preferences of the new device to mimic the look and feel of the old device. In addition, by using a peer-to-peer connection and not relying on other communication methods like Wi-Fi, the setup of the new device may take place even when Internet connectivity is not possible or is limited in speed or bandwidth.

(5) Details of one or more implementations are set forth in the accompanying drawings and the description below. Other features, aspects, and potential advantages will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

(1) FIGS. 1A-1B illustrate block diagrams of example systems for migrating data from a source device to a destination device for initial setup of the destination device.

(2) FIGS. 2A-2B show example user interfaces (UIs) that may be used to migrate data from a source device to a destination device.

(3) FIG. 3 shows an example system interaction diagram for migrating data from a source device to a destination device.

- (4) FIG. 4 shows an example system for providing a second reboot without user interaction.
- (5) FIG. 5 is a flow diagram of an example process for configuring a destination device based on a current state of a source device using a peer-to-peer connection.
- (6) FIG. 6 is a flow diagram of an example process for configuring a destination device based on a current state of a source device using a peer-to-peer connection.
- (7) FIG. 7 is a flow diagram of an example process for configuring a destination device based on a current state of a source device using a peer-to-peer connection.
- (8) FIG. 8 is a block diagram of an example computing device that can implement the features and processes of FIGS. 1A-7.
- (9) Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

- (10) Examples provided herein add context and aid in the understanding of the described embodiments. It will be apparent to one skilled in the art that the described embodiments may be practiced without some or all of these specific details or performed in a different order to obtain the same effect. In other instances, well known process steps may not be described in detail in order to avoid convoluting the described embodiments. Other applications are possible, such that the following examples should not be taken as limiting.
- (11) The embodiments described herein set forth techniques for enabling data migration between a source device (e.g., old device) and a destination device (e.g., new device) for initially setting up the destination device. According to some embodiments, the data migration may be carried out over a private wireless network established between the source device and destination device, over a wireless peer-to-peer connection, over a physical wired connection, or via a direct connection. Accordingly, the techniques described herein provide a mechanism for migrating data between the source and destination devices that are owned by a user in a secure manner. A more detailed discussion of these techniques is set forth below.
- (12) FIG. 1A illustrates a block diagram of an example system **100** for migrating data from a source device **102** (e.g., old device) to a destination device **104** (e.g., new device) for initial setup of the destination device **104**. In one approach, each of source device **102** and destination device **104** may be owned and/or operated by a same user (e.g., associated with the same user account as described later), but embodiments set forth herein are not limited to such a scenario. Source device **102** may be any type of mobile computing device, such as a smartphone, mobile phone, tablet computer, notebook computer, laptop computer, etc., that runs a first operating system (OS). Destination device **104** may be any type of mobile computing device running a second OS. Moreover, destination device **104** may be a computing device of the same type or different type as source device **102**, in an example.
- (13) According to an example, destination device **104** may be a new smartphone that has been recently purchased or recently acquired by a user and source device **102** may be the user's old smartphone that the user owns and has been operating over time, such as a generation older smartphone. The user, in this example, may desire to transfer all the user's data from the old smartphone to the new smartphone such that the transition from the old smartphone to the new smartphone is seamless for the user.
- (14) The data that is migrated from source device **102** to destination device **104** may include, for example, contact information, calendar information, messages and message history, emails, bookmarks, wallpaper settings, information regarding installed applications, photos, videos, songs, movies, books, other media, personal data, messages, social media posts, files, documents, device and application settings, device and application preferences, applications, and/or other irreplaceable data.
- (15) In another example, source device **102** may be connected directly to destination device **104** via direct connection **106**. Direct connection **106** may be a hardwire connection in an example.
- (16) In another example, source device **102** may be connected to destination device **104** via a

wireless connection **108**. Any type of wireless connection **108** may be used, such as Bluetooth, peer-to-peer network, near-field communication (NFC), ultra wideband (UWB), etc.

(17) These peer-to-peer connections **106**, **108** may be used to transfer data from source device **102** to destination device **104** in order to setup and/or configure destination device **104** in a manner similar to how source device **102** is setup and/or configured. This similarity may take into account any underlying differences between devices, such as different operating systems, memory capacity, display size, native applications installed, etc.

(18) Although not shown in FIG. 1A, processor **110**, in conjunction with memory **112**, may execute and/or provide first OS that is configured to execute various applications (e.g., native OS applications and/or user applications) on source device **102**. Similarly, processor **114**, in conjunction with memory **116**, may execute and/or provide second OS that is configured to execute various applications (e.g., native OS applications and user applications) on destination device **104**. Second OS may be the same or different from first OS, and any compatibility issues between the various operating systems may be overcome using techniques described herein.

(19) FIG. 1B illustrates a block diagram of another example system **118** for migrating data from source device **102** (e.g., old device) to destination device **104** (e.g., new device) for initial setup of destination device **104**. In an example, source device **102** and destination device **104** may be configured to communicate via a network **120**. In one approach, network **120** may be hosted by source device **102**, by destination device **104**, or by some other device or system. In another approach, network **120** may be used to migrate the data from source device **102** to destination device **104**.

(20) Any type of network **120** may be utilized, such as the Internet, wide area network (WAN), local area network (LAN), wireless local area network (WLAN), virtual private network (VPN), Bluetooth network, Bluetooth low energy (BTLE) network, peer-to-peer network, mobile broadband network, etc. Moreover, in an approach, more than one network may be used to connect source device **102** and destination device **104**. Source device **102** and destination device **104** may be wirelessly coupled to network **120** via wireless connections **124**, **126**, respectively. Wireless connections **124**, **126** may utilize any protocol known in the art to communicate through network **120** with the other device.

(21) In one example, if an Internet or network connection is available to source device **102**, then synchronization of any user data of source device **102** may be determined prior to choosing how and which data to migrate to destination device **104**. Should some first user data be synchronized with cloud server **122**, then the user may choose to not have the first user data transferred from source device to destination device **104**, to save time and conserve processor workload and network bandwidth. In another example, source device **102** may automatically select to not have the first user data transferred from source device to destination device **104** when a network connection is available to conserve resources.

(22) In addition to transferring data that is not synchronized with cloud server **122**, the first user data may be downloaded and/or synchronized to destination device **104**, thereby causing the first user data to be stored or synchronized to destination device **104** and available for use with any user applications, settings, preferences, etc. However, it is not necessary that an Internet or network connection be utilized to configure and setup destination device **104**, as all necessary data may be obtained from source device **102** via the secure peer-to-peer connection(s) **106**, **108**.

(23) In another example, a cloud server **122** may be used in the migration process. Cloud server **122** may, in an example, host a migration application that provides migration of the data from source device **102** to destination device **104**. However, migration of the data that takes place without the need for Internet or network connectivity of source device **102** and destination device **104** may be preferred to allow the data migration even when a network **120** is unavailable or congested, in an approach. In addition, some data from source device **102** may be stored and/or backed-up on cloud server **122** prior to migration to destination device **104**. For example, photos

and videos that have been captured using source device **102** may be automatically backed-up to cloud server **122** and associated with a user account. This technique ensures that the user does not lose this content should the user lose or damage source device **102**. Moreover, after the data migration and setup processes are performed, destination device **104** may be associated with the user account, providing the user with access to the content stored to cloud server **122** through destination device **104**.

(24) To enhance the user's privacy, any data that is sent to network **120** and/or cloud server **122** may be encrypted to ensure that unauthorized third parties are not able to obtain the data. Moreover, the user may be requested to provide permission for any transmission of data to network **120** and/or cloud server **122** prior to such a transmission taking place, such as by showing a pop-up window or notification requesting the user's permission to send the data. Should the user not provide permission to send the data, the process may be suspended until a secure peer-to-peer connection is available.

(25) FIGS. 2A-2B show example user interfaces (UIs) that may be used to migrate data from a source device **102** to a destination device **104**. In FIG. 2A, example data transfer screens **202**, **204** are shown on source device **102**. Data transfer screens **202**, **204** may include text, graphics, video, images, and/or user interactive elements, such as buttons, toggles, sliders, radio buttons, etc. In one approach, a data migration process may be initialized or started from data transfer screen **202**, such as by the user providing input to start the migration process by selecting a "Yes" button in response to being asked if the user wants to migrate data to destination device (e.g., the new device).

(26) In one approach, a QR code **210** may be generated by source device **102** and presented on a data transfer screen **204** of source device **102** along with instructions for the user to scan the QR code **210** with an optical sensor (e.g., camera) of the destination device (e.g., new device). In another approach, a code may be presented on data transfer screen **204** of source device **102** along with instructions for the user to enter the code at an appropriate place on a setup screen of the destination device.

(27) As will be appreciated, other forms of codes and/or pairing information (e.g., an alphanumeric code, barcodes, pairing images, etc.) may be generated by source device **102** or the destination device for purposes of establishing a secure connection (e.g., a private wireless network, NFC connection, BTLE connection, UWB connection, etc.) with the other device. In a further approach, the secure connection may be a secure peer-to-peer connection as described herein or of a type known in the art.

(28) In response to the user performing the instructed action(s) (e.g., scanning the QR code **210**, entering a code, scanning an image or series of images, etc.), source device **102** or the destination device may establish the secure peer-to-peer connection with the other device. In some embodiments, source device **102** may disconnect any existing wireless network connections prior to establishing the secure peer-to-peer connection. In some embodiments, source device **102** may serve as a wireless access point and function as a hotspot, such that a private wireless network hosted by source device **102** may be utilized for data exchange between source device **102** and the destination device.

(29) Source device **102** may derive, based on the code entered or scanned at the destination device, a number of connection parameters associated with the secure peer-to-peer connection, such as a connection type, connection protocol(s), network name, also referred to as a service set identifier (SSID), pre-shared key(s) (or other shared secret information) used for encryption purposes, etc. In some embodiments, source device **102** may initiate a number of services associated with the secure peer-to-peer connection. For example, the services may include Dynamic Host Configuration Protocol (DHCP) for Internet Protocol (IP) address allocation, multicast Domain Name System (mDNS) for discovery of source device **102** and destination device **104**, and/or other services known in the art for use in establishing and maintaining hardwired or wireless secure peer-to-peer connections.

(30) In another example, a message or handshake exchange may take place between source and destination devices, which may include a series of keys and/or certificates (that offer an additional layer of security at transport layer) to ensure that any communication over the secure peer-to-peer connection will be performed securely and without data loss. In some embodiments, during the handshake exchange, the source device and destination device **104** may exchange basic information regarding capabilities of the devices (e.g., wireless protocol(s) supported, hardware features, information transfer rate, types of data to be migrated (e.g., photos, videos, etc.)). Moreover, the source device and destination device **104** may negotiate an appropriate wireless protocol to be used for the data migration process based on the exchange.

(31) In one approach, the QR code **210** may be generated in response to selection of the “Yes” button on the data transfer screen **202**. In some embodiments, when the user selects the “Yes” button, source device **102** may perform one or more operations. The operations may include, in an example: 1) generating the code, 2) presenting the code on data transfer screen **202**, 3) deriving parameters to provide the secure peer-to-peer connection, 4) establishing the secure peer-to-peer connection, and 5) initiating services associated with the secure peer-to-peer connection. At this point, source device **102** and the destination device are not in communication with one another via the secure peer-to-peer connection.

(32) In FIG. 2B, example data transfer screens **206**, **208** are shown on destination device **104**. Data transfer screens **206**, **208** may include text, graphics, video, images, and/or user interactive elements, such as buttons, toggles, sliders, radio buttons, etc. In one approach, a data migration process may be initialized or started from setup screen **206**, such as by the user providing input to start the migration process by selecting a “Yes” button in response to being asked if the user wants to setup destination device **104** with data from the source device (e.g., another device).

(33) In one approach, the QR code **210** may be scanned by the user using an optical sensor (e.g., a camera) of destination device **104** using scanning window **212**. In another approach, a code may be presented on the source device to be input into a text box or some other appropriate place on a setup screen of destination device **104**. As will be appreciated, other forms of codes and/or pairing information may be generated by the source device or destination device **104** for purposes of establishing a secure peer-to-peer connection with the other device.

(34) In response to the user performing the instructed action(s) (e.g., scanning the QR code **210**, entering a code, scanning an image or series of images, etc.), destination device **104** or the source device may establish the secure peer-to-peer connection with the other device. In some embodiments, destination device **104** may disconnect any existing wireless network connections prior to establishing the secure peer-to-peer connection. In some embodiments, destination device **104** may serve as a wireless access point and function as a hotspot, such that a private wireless network hosted by destination device **104** may be utilized for data exchange between destination device **104** and the source device. Destination device **104** may derive, based on the code entered or scanned at the destination device, a number of connection parameters associated with the connection type. In some embodiments, destination device **104** may initiate a number of services associated with the secure peer-to-peer connection.

(35) In one approach, the scanning window **212** may be provided on setup screen **208** in response to selection of the “Yes” button on the setup screen **206**. In some embodiments, when the user selects the “Yes” button, destination device **104** may perform one or more operations. The operations may include, in an example: 1) providing the scanning window **212** or text box for code entry, 2) deriving parameters to provide the secure peer-to-peer connection, 3) establishing the secure peer-to-peer connection, and 4) initiating services associated with the secure peer-to-peer connection. At this point, destination device **104** and the source device are not in communication with one another via the secure peer-to-peer connection.

(36) In one approach, once the QR code **210** is scanned and verified by destination device **104**, the setup screen **208** may display instructions that explain operations to be carried out by the user at

destination device **104**. The operations may include, in an example: 1) access an online application store, 2) download a migration application that facilitates the migration of data from source device to the destination device **104**, and 3) open or start the migration application.

(37) In another approach, the migration application may be included with the first and/or second OS, such that steps **1** and **2** above do not need to be performed to start the migration application. In some embodiments, the migration application may be provided by a manufacturer of the source device or destination device **104**.

(38) Once the user initiates the data migration process, either from the source device or destination device **104**, in an approach, no further input is necessary from the user to cause all data to be migrated to destination device **104**.

(39) FIG. **3** shows an example system interaction diagram for migrating data from a source device to a destination device. The interactions between source device **102** and destination device **104** may be performed in response to a secure peer-to-peer connection being established between source device **102** and destination device **104**. In an approach, destination device **104** may receive a request **302** to migrate data from source device **102** via the peer-to-peer connection. This request **302** may be received by user input to a UI displayed to destination device **104** in one example. In another example, the request **302** may be sent from source device **102** to destination device **104** in response to user input to source device **102** indicating a desire to migrate data to destination device **104**.

(40) In one example, all user data may be transferred from the user's old smartphone to the user's new smartphone. This transfer may take place at the place of purchase of the new smartphone (e.g., mobile telephone store, kiosk, etc.), at home, or any place desired by the user where the old smartphone and new smartphone are in proximity with one another and under the user's control. Moreover, system files and OS specific data may remain on the old smartphone and not be transferred to the new smartphone, since such files and data may be unusable by the OS of the new smartphone, and/or would not be needed to reconstruct the look and feel of the old smartphone.

(41) In an approach, destination device **104** may send an initialization message **304** to source device **102**. The initialization message **304** may include any information that is useful in prompting the receiving device to prepare for a data migration and identify the sending device to ensure that it is authorized to receive the data transfer. In one approach, initialization message **304** may include device details about destination device **104**, such as a device ID, OS type and version, class or type of the device (e.g., smartphone, tablet, laptop, etc.), device model (specific manufacturer and model number of the device), marketing name of the device, data classification to check for compatibility, etc.

(42) In this approach, source device **102** may review all of the information provided by destination device **104** to ensure that a data transfer will be successful and is authorized to go forward, based on compatibility with OS, data class, formats, and file structures of destination device **104**. When the review is positive, source device **102** may respond to the initialization message **304** with an initialization response **306**. The initialization response **306** may acknowledge receipt of the initialization message **304** and indicate device information for source device **102**.

(43) According to another approach, the initialization message **304** may be sent from source device **102** to destination device **104**, and the initialization response **306** may be sent from destination device **104** to source device **102**. The information included in each message will be particular to the device which sends the message/response.

(44) In response to receiving the initialization response **306**, destination device **104** may send a preflight message **308** to source device **102**. The preflight message **308** may include information specific to the data transfer as known by destination device **104** and the capabilities of the destination device **104**, such as an amount of available memory on destination device **104**, a manifest describing a file format and structure for destination device **104**, etc.

(45) According to an approach, in response to receiving the initialization message **304** and/or the

preflight message **308**, source device **102** may generate, in operation **326**, a snapshot of the full file system of source device **102** which includes all data that is stored to source device **102** at a time that the snapshot is taken. For example, the snapshot may record an image of the file system of source device **102** that shows the existence of every file on source device **102** and indicates contents of each file. The image may, in one example, include a serialized copy of the entire state of source device **102** stored in a non-volatile form, such as a file. Destination device **104** may utilize the image to reconstruct the system of source device **102** by shutting down and rebooting to exactly the same state as source device **102**. Any method known in the art may be used to create the snapshot as would be understood by one of skill in the art.

(46) Once the snapshot is generated at source device **102**, in an approach, source device **102** may be locked to ensure that no further changes are made to data on the device prior to being transferred to destination device **104**. In approaches where source device **102** is not locked, incremental snapshots that indicate modified content that occurred after the snapshot was generated may be made to account for any changes that occur post-snapshot.

(47) In another approach, in response to receiving the initialization message **304** and/or the preflight message **308**, source device **102** may prepare a manifest that describes a set of data stored to source device **102** that will be transferred to destination device **104**, along with locations associated with the set of data to be used in moving the set of data to its corresponding storage location after it is received on destination device **104**. In one implementation, the set of data may include all user data based on the snapshot, and may exclude files and data that are specific to the first OS operating on source device **102** and other data that is unrelated to the user's interactions with source device **102**. There are many reasons why only user data is transferred to destination device **104**. Some reasons include that the excluded system data may not be applicable to the second OS operating on destination device **104**, may be unreadable by destination device **104**, may represent files and data that would be duplicated on destination device **104**, etc. Therefore, to avoid these situations and the unnecessary use of bandwidth and processing capacity to move the system files, they are excluded from the set of data transferred to destination device in this approach.

(48) Source device **102**, upon receiving the preflight message **308**, may respond with a preflight response **310**. The preflight response **310** may include information specific to the data transfer as known by source device **102** along with capabilities of the source device **102**, such as a number of files that will be transferred to destination device **104**, a total number of settings and/or preferences that will be transferred to destination device **104**, which aspects of the device the settings and/or preferences apply to, etc. This information may be used to determine progress of the data transfer by destination device **104** once the data begins to be transferred.

(49) According to another approach, the preflight message **308** may be sent from source device **102** to destination device **104**, and the preflight response **310** may be sent from destination device **104** to source device **102**. The information included in each message will be particular to the device which sends the message/response.

(50) Upon receiving the preflight response **310**, in one approach, source device **102** may begin transferring data in a series of packages or uploads (e.g., upload #1, upload #2, . . . upload #n, collectively **314**) until all the data from the manifest is transferred to destination device **104**. In this approach, upon receipt of each upload **314**, destination device may store **316** the data within the particular upload to a staging area or backup store.

(51) According to some embodiments, the data migration of the successive uploads **314** from source device **102** to destination device **104** may be fault-tolerant. In other words, if the secure peer-to-peer connection is lost or one of the devices loses power (due to reboot or other reasons), the data migration process is only temporarily interrupted and may be resumed as soon as the secure peer-to-peer connection or device power is restored. In some implementations, a status of the data migration process may be maintained on source device **102**, on destination device **104**, or on both. The device that maintains the status of the migration process may keep a record of a

particular state in the data migration process. The particular state may include information regarding the operation that was interrupted (e.g., upload #2, upload #n, transfer of manifest, etc.) that was being performed when the connection or device power was lost, an amount of data (e.g., bytes of data, upload number in the sequence, a number of records transferred) that was migrated or transferred prior to losing the connection or device power.

(52) In some embodiments, the data migration may be paused and resumed as desired by a user without affecting the consistency of the migration process. In some embodiments, source device **102** and destination device **104** may be usable while the data migration process is being carried out. Also, because the data migration occurs over the secure peer-to-peer connection, which may be hosted by source or destination device and is not dependent on an external network environment the devices are in, the devices may continue the data migration process (so long as the secure peer-to-peer connection between the devices remains intact). For example, if a user is at a retail store where the data migration process is started over the secure peer-to-peer connection, the user can walk out of the retail store with the two devices without interrupting the data migration process as the process does not rely on any third-party network (e.g., retail store wi-fi network).

(53) In some embodiments, when an existing secure peer-to-peer connection between source device **102** and destination device **104** is disconnected during the data migration process and the user attempts to restart the data migration process, a different secure peer-to-peer connection may be established based on a new authentication exchange (e.g., code, image, etc.) generated at the initiating device. The other device may derive the connection parameters associated with establishing the new secure peer-to-peer connection. When source and destination devices discover one another, they are able to identify one another (e.g., based on unique identifiers), and the data migration process may be resumed from a point where the previous secure peer-to-peer connection was disconnected/terminated.

(54) In an approach, source device may send the manifest **318** to destination device **104** for use in relocating the stored data from the staging area/backup store to corresponding locations indicated in the manifest. The manifest may be sent at any time after it is generated in operation **328**, such as being sent with or just after the preflight response **310**, being sent before or after one of the uploads **314**, or being sent once all data has been transferred from source device **102** to destination device **104**.

(55) Optionally, after the manifest **318** is sent, source device **102** may prompt destination device **104** to begin processing the transferred data and/or metadata associated with the transferred data by sending a relocate message **320** indicating that no further data will be sent from source device **102**. This information may prompt destination device **104** to reboot, and then begin unpackaging, decompressing, decrypting, analyzing, and/or relocating the transferred data and/or metadata associated with the transferred data it has already received. In one approach, when the manifest **318** is sent after all the data uploads **314** are complete, the transmission of the manifest **318** may signal the end of data transmission, in which case the relocate message **320** is unneeded.

(56) In one approach, the set of data transferred to destination device **104** may be compressed on source device **102** prior to being transferred. Once it is uploaded to destination device **104**, a decompression process may be performed to the transferred data to decompress it for use on destination device **104**. Any compression algorithm known in the art may be used by source device **102**, while any corresponding decompression algorithm known in the art may be used by destination device **104** to compress/decompress the set of data. This process of compression/decompression reduces processor and bandwidth requirements to perform the data transfer.

(57) In operation **322**, destination device reboots, then after rebooting it relocates the stored data and/or metadata associated with the stored data to corresponding locations specified by the manifest. Moreover, if the data was sent distributed across several packages and/or encrypted, destination device **104** may process the received data to reassemble the packages and/or decrypt the

data prior to relocating the data to its corresponding location on destination device **104**.

(58) In one approach, the transferred data may be cloned on destination device **104** to its corresponding locations by moving and/or copying metadata associated with the transferred data to the corresponding locations specified by the manifest, while maintaining a storage location of the data itself. This process of cloning reduces processor cycles used, and expedites setup of destination device **104** for faster utilization by the user.

(59) Once all data has been processed by destination device **104**, it may send a completion message **324** to source device **102** indicating that the process has ended. Moreover, this completion message **324** may prompt one or both devices to terminate the secure peer-to-peer connection therebetween.

(60) In an example, any user accounts that the user had established with or accessed with source device **102** may be automatically re-established on destination device **104**, so that the user may enjoy all the features and functionality afforded to the user through use of the specific user accounts. Any number of user account(s) may be used in this fashion, and may include accounts for online music streaming services, online backup services, online video streaming services, online storage services, etc. The re-establishment of the user account(s) may provide destination device **104** with access to data stored online in an approach. Moreover, this data that is accessible online may be removed from the data transferred to destination device **104** from source device **102** in anticipation of later gaining this access through the user account(s) being established on destination device **104** automatically.

(61) In response to the completion message **324** being received by source device **102**, a selectable notification may be presented on source device **102** asking the user if the device should be wiped and/or all user data deleted and returned to a default state. If the user selects the notification and chooses to wipe source device **102**, a removal process may be started to return source device **102** to a default state and remove all user data.

(62) In one approach, this removal process that returns source device **102** to the default state that removes all user data may be performed automatically once the completion message **324** is received on source device **102**. In a further approach, the user may be asked whether to wipe source device **102** after completing the migration process at a start of the migration process, and will proceed according to the user's previous instructions received at the beginning of the migration process.

(63) Optionally, once all data has been unpackaged, decompressed, decrypted, analyzed, and/or relocated as described by the manifest, destination device **104** may be rebooted again to a lock screen for the user to enter credentials to login to the destination device which has been setup to have all the user data, settings, preferences, etc., of source device **102**.

(64) FIG. 4 shows an example system **400** for providing a second reboot without user interaction. The second reboot may be possible by stopping all daemons operating on the destination device except for one or more essential daemons that keep the kernel running **422**, then restarting all the other non-essential daemons with the new configuration specified by the manifest and current state of source device. Many times an OS may restrict a process (such as the data migration process) to only rebooting after receiving user interaction. However, in the present descriptions, two reboots are used. This second reboot is provided as a software reboot that strips down all non-essential processes while keeping the kernel running.

(65) This technique for providing a second reboot does not require the user to login to the device until the full migration process has completed and the device is ready to be used again. Because the initial transfer of data from the source device to the destination device may take place over a first period of time, while the relocating of the data once transferred to the destination device may take place over a second time period, it is undesirable for the user to be required to interact with the destination device before transitioning from uploading the data to relocating the data.

(66) Example system **400** shows user space processes **420** that are operated above the OS level **428**. The kernel **422** is maintained in the OS level **428** during the second reboot, but any non-

essential daemons, processes, or services from the user space processes **420** are terminated to allow the migrated data to be configured appropriately. For example, user manager daemon **404**, window manager daemon **414**, display serve daemon **416**, and background processes daemon **418** may all be terminated during the second reboot. Moreover, any secondary process that relies on a terminated daemon, such as the user session manager **406** may also be terminated, along with any applications managed by a terminated secondary process, such as the login interface **408**, user interfaces **412**, and user applications **410**.

(67) According to an example, some OS level **428** processes may continue and some may be terminated when performing the second reboot. In this example, I/O device management **424** may be terminated during the second reboot, while security services **426** may remain running to provide security and privacy services for the destination device.

(68) Moreover, access to the staging area memory **432** will remain unaffected by the second reboot in an example, while access to some other areas of the memory system **430** may be suspended due to termination of one or more processes, daemons, or services in the OS level **428**.

(69) In an example, if two game applications and their associated data (e.g., last saved position in the game, equipment collected, special maps, etc.) are transferred from the source device to the destination device, then the user manager daemon **404** would be terminated on the destination device so that the applications and associated data may be loaded and configured to operate as they did on the source device. After the second reboot, the user manager daemon **404** would be restarted and these game applications would be ready to be played by the user from the same place that they were last played on the source device.

(70) In another example, if a custom desktop is used on the source device, then after migrating over the data for this custom desktop to the destination device, the user interface process **412** would be terminated along with the user manager daemon to allow for the custom desktop to be installed on the destination device. After the second reboot, the user manager daemon **404** would be restarted and the destination device would be operable using the custom desktop just like on the source device.

Example Processes

(71) To enable the reader to obtain a clear understanding of the technological concepts described herein, the following processes describe specific steps performed in a specific order. However, one or more of the steps of a particular process may be rearranged and/or omitted while remaining within the contemplated scope of the technology disclosed herein. Moreover, different processes, and/or steps thereof, may be combined, recombined, rearranged, omitted, and/or executed in parallel to create different process flows that are also within the contemplated scope of the technology disclosed herein. Additionally, while the processes below may omit or briefly summarize some of the details of the technologies disclosed herein for clarity, the details described in the paragraphs above may be combined with the process steps described below to get a more complete and comprehensive understanding of these processes and the technologies disclosed herein.

(72) FIG. 5 is a flow diagram of an example method **500** for configuring a destination device based on a current state of a source device using a peer-to-peer connection. In operation **502**, the destination device may receive input to configure the destination device based on a current state of the source device. The current state of the source device may be determined based on a snapshot taken by the source device at a particular point in time.

(73) The input may specify that the user desires for the destination device to be set up according to how the source device is currently operating, e.g., settings, preferences, applications installed, data and files stored to memory, etc. This input may be received via a UI displayed by the destination device, in one approach. Moreover, the input may be touch input received via a touchscreen, voice input received via a microphone, motion input received via a motion capture device, or some other known input device known in the art, such as keyboard, mouse, etc.

(74) In operation **504**, the destination device may pair the source device with the destination device in response to the input. This pairing establishes a secure peer-to-peer connection as described herein, for transferring data from the source device to the destination device and exchanging messages therebetween.

(75) The secure peer-to-peer connection may be established using any technique described herein and others known in the art, such as entry of a code, display of an image, series of images, barcode, UPC, QR code, etc., on one device and capturing this information using the other device, etc.

(76) In several approaches, the destination device may be connected to the source device, prior to transferring data objects, via a Wi-Fi peer-to-peer connection, via a physical wired connection, or via a direct connection.

(77) In operation **506**, the destination device may transfer (e.g., pull) a plurality of data objects from the source device to be stored to the destination device. The destination device effectively pulls the data objects from the source device without the source device being involved in the transfer operation, e.g., requesting specific data objects for transfer, accessing memory of the source device and copying data objects therefrom, or some other known way of pulling data from another device. In another approach, source device may transfer (e.g., push) the data objects to the destination device.

(78) In one approach, destination device may determine whether there is sufficient available memory on the destination device prior to beginning the transfer. Otherwise, method **500** may end without any data being transferred.

(79) The data objects are selected from the snapshot of the current state of the source device at a current point in time when the snapshot is taken. Data objects may include files, settings, preferences, images, photos, videos, messages, account information, and any other user data stored to the source device.

(80) In one approach, system and OS-specific data may be excluded from the data objects selected from the snapshot. In this approach, the data objects may include settings, preferences, configurations, photos, music, videos, messages, applications, and other user data. In one approach, user data may include financial information, like particular online and electronic payment methods and information that may be used to purchase goods and services using source device **102**. In addition, system files of the source device may be excluded from the data objects.

(81) In operation **508**, the destination device may store the data objects in a first storage location on the destination device. This first storage location may be a staging area or backup store configured to temporarily store data for further processing.

(82) In operation **510**, the destination device may transfer (e.g., pull) a manifest from the source device to be stored to the destination device. The manifest includes information describing the data objects and relative locations of the data objects on the source device for use in relocating the data objects to corresponding storage locations once stored to the first storage location on the destination device. In one approach, the manifest may include indication of a total number of the data objects that will be transferred from the source device to the destination device.

(83) In operation **512**, the destination device may be rebooted into a configuration mode in response to the input to configure the destination device. The configuration mode allows for exclusive access to a storage medium of the destination device, such as the first storage location. The configuration mode is entered after rebooting the destination device, without user interaction causing the destination device to reboot or enter the configuration mode.

(84) In operation **514**, while in the configuration mode, the destination device may move each of the data objects in the first storage location to second storage locations corresponding to their respective relative locations indicated in the manifest. The movement of the data objects may take place because in configuration mode, the process executing the movement commands has exclusive access to memory of the destination device.

(85) In operation **516**, the destination device may reboot in response to completing moving the data

objects to the second locations. This allows for the destination device to be configured to access the data objects in their respective relative locations. This second reboot may also take place without user interaction, such as logging in, providing credentials, etc., thereby making the whole migration process seamless for the user with only the first interaction being needed to start the process.

(86) In an approach, method **500** may include determining, by the destination device, whether an OS update is available for the destination device. For example, the OS may have a security patch, feature enhancement, or some other update that has not been performed but is available for the user. In response to determining that the update is available, the destination device may update the first OS of the destination device prior to transferring the data objects.

(87) In one implementation, method **500** may include establishing, by the destination device, a synchronization process between a remote or cloud storage and at least some of the data objects that were previously synchronized with the remote or cloud storage on the source device. In this implementation, after transferring over the data objects that were synchronized with the remote or cloud storage, the automatic backup to the remote or cloud storage may be re-established to ensure that any changes to the synchronized data objects will be captured in the back-up location.

(88) According to an example, method **500** may include determining, by the destination device, a size of the data objects. This size is a measurement of how much storage space is required to store the data objects on the destination device. Further, method **500** may include determining, by the destination device, whether an amount of available storage on the destination device is sufficient to store the data objects to the destination device. In this example, the destination device may transfer the data objects from the source device in response to a determination that the amount of available storage on the destination device is sufficient to store the data objects to the destination device, and the destination device may not transfer any data from the source device in response to a determination that the amount of available storage on the destination device is insufficient to store the data objects.

(89) FIG. **6** is a flow diagram of an example method **600** for configuring a destination device based on a current state of a source device using a peer-to-peer connection. In operation **602**, the source device may receive input to configure the destination device based on a current state of the source device. The current state of the source device may be determined based on a snapshot taken by the source device at a particular point in time.

(90) The input may specify that the user desires for the destination device to be set up according to how the source device is currently operating, e.g., settings, preferences, applications installed, data and files stored to memory, etc. This input may be received via a UI displayed by the source device, in one approach. Moreover, the input may be touch input received via a touchscreen, voice input received via a microphone, motion input received via a motion capture device, or some other known input device known in the art, such as keyboard, mouse, etc.

(91) In operation **604**, the source device may pair with the destination device in response to the input. This pairing establishes a secure peer-to-peer connection as described herein, for transferring data from the source device to the destination device and exchanging messages therebetween.

(92) The secure peer-to-peer connection may be established using any technique described herein and others known in the art, such as entry of a code, display of an image, series of images, barcode, UPC, QR code, etc., on one device and capturing this information using the other device, etc. In several approaches, the source device may be connected to the destination device, prior to transferring data objects, via a Wi-Fi peer-to-peer connection, via a physical wired connection, or via a direct connection.

(93) In operation **606**, the source device may capture a snapshot of the current state of the source device at a current point in time. Any snapshot technique known in the art may be used by the source device.

(94) In one approach, the source device may be locked after capturing the snapshot to prevent user interaction with the source device. In this approach, the source device may remain locked until the

data objects are transferred to the destination device.

(95) In operation **608**, the source device may select a plurality of data objects from the snapshot to be transferred to the destination device. The selected data objects correspond to the current state of the source device at the current time, and may include all user data on the source device. Moreover, in an example, data objects may include files, settings, preferences, images, photos, videos, messages, account information, and any other user data stored to the source device.

(96) In one approach, system and OS-specific data may be excluded from the data objects selected from the snapshot. In this approach, the data objects may include settings, preferences, configurations, photos, music, videos, messages, applications, and other user data. In addition, system files of the source device may be excluded from the data objects.

(97) In operation **610**, a manifest may be generated by the source device. The manifest includes information describing the data objects and relative locations of the data objects on the source device for use in relocating the data objects to corresponding storage locations once stored to the destination device. In one approach, the manifest may include indication of a total number of the data objects that will be transferred from the source device to the destination device.

(98) In an approach, the source device may determine a total number of the data objects, and this number may be stored to the manifest. In this way, the destination device may be able to track a progress of the data transfer using the total number of data objects to be transferred, and be able to determine when the transfer is complete.

(99) In operation **612**, the source device may transfer (e.g., push) the data objects from the source device to the destination device. In another approach, the destination device may transfer (e.g., pull) the data objects from the source device to be stored to the destination device.

(100) In one approach, source device may determine whether there is sufficient available memory on the destination device prior to beginning the transfer. Otherwise, method **600** may end without any data being transferred.

(101) In operation **614**, the source device may transfer the manifest from the source device to the destination device. The manifest may be used by the destination device to know where to move each of the data objects after being stored initially to a staging area or backup store in order to reproduce the environment of the source device, e.g., to relocate each data object to storage locations corresponding to their respective relative locations indicated in the manifest.

(102) In an approach, method **600** may include the source device determining whether an OS update is available for the destination device. For example, the OS may have a security patch, feature enhancement, or some other update that has not been performed but is available for the user. In response to determining that the update is available, the destination device may be requested to update the first OS of the destination device prior to transferring the data objects.

(103) In one implementation, method **600** may include the source device providing indication to the destination device of at least some of the data objects that were previously synchronized with a remote or cloud storage on the source device. In this way, the destination device may establish a synchronization process between the remote or cloud storage and the data objects that were previously synchronized with the remote or cloud storage on the source device after the transfer.

(104) In another approach, the source device may compress the data objects prior to transferring the data objects to the destination device, using any known compression algorithm.

(105) According to an example, method **600** may include determining, by the source device, a size of the data objects. This size is a measurement of how much storage space is required to store the data objects on the destination device. Further, method **600** may include determining, by the source device, whether an amount of available storage on the destination device is sufficient to store the data objects to the destination device. In this example, the source device may transfer the data objects to the destination device in response to a determination that the amount of available storage on the destination device is sufficient to store the data objects to the destination device, and the source device may not transfer any data to the destination device in response to a determination that

the amount of available storage on the destination device is insufficient to store the data objects. (106) FIG. 7 is a flow diagram of an example method **700** for configuring a destination device based on a current state of a source device using a peer-to-peer connection. In operation **702**, input is received to configure the destination device based on a current state of the source device. The current state of the source device may be determined based on a snapshot taken by the source device at a particular point in time.

(107) The input may specify that the user desires for the destination device to be set up according to how the source device is currently operating, e.g., settings, preferences, applications installed, data and files stored to memory, etc. This input may be received via a UI displayed by the source or destination device, in one approach. Moreover, the input may be touch input received via a touchscreen, voice input received via a microphone, motion input received via a motion capture device, or some other known input device known in the art, such as keyboard, mouse, etc.

(108) In operation **704**, the source device and the destination device may be paired in response to the input. This pairing establishes a secure peer-to-peer connection as described herein, for transferring data from the source device to the destination device and exchanging messages therebetween.

(109) The secure peer-to-peer connection may be established using any technique described herein and others known in the art, such as entry of a code, display of an image, series of images, barcode, UPC, QR code, etc., on one device and capturing this information using the other device, etc. In several approaches, the source device may be connected to the destination device, prior to transferring data objects, via a Wi-Fi peer-to-peer connection, via a physical wired connection, or via a direct connection.

(110) In operation **706**, the source device may capture a snapshot of the current state of the source device at a current point in time. Any snapshot technique known in the art may be used by the source device.

(111) In one approach, the source device may be locked after capturing the snapshot to prevent user interaction with the source device. In this approach, the source device may remain locked until the data objects are transferred to the destination device.

(112) In operation **708**, the source device may select a plurality of data objects from the snapshot to be transferred to the destination device. The selected data objects correspond to the current state of the source device at the current time, and may include all user data on the source device. Moreover, in an example, data objects may include files, settings, preferences, images, photos, videos, messages, account information, and any other user data stored to the source device.

(113) In one approach, system and OS-specific data may be excluded from the data objects selected from the snapshot. In this approach, the data objects may include settings, preferences, configurations, photos, music, videos, messages, applications, and other user data. In addition, system files of the source device may be excluded from the data objects.

(114) In operation **710**, a manifest may be generated by the source device. The manifest includes information describing the data objects and relative locations of the data objects on the source device for use in relocating the data objects to corresponding storage locations once stored to the destination device. In one approach, the manifest may include indication of a total number of the data objects that will be transferred from the source device to the destination device.

(115) In an approach, the source device may determine a total number of the data objects, and this number may be stored to the manifest. In this way, the destination device may be able to track a progress of the data transfer using the total number of data objects to be transferred, and be able to determine when the transfer is complete.

(116) In operation **712**, the data objects may be transferred to the destination device, using a pull operation, a push operation, or a combination of pushing from the source device and pulling from the destination device.

(117) In one approach, source device may determine whether there is sufficient available memory

on the destination device prior to beginning the transfer. Otherwise, method **700** may end without any data being transferred.

(118) In operation **714**, the destination device may store the data objects to a first storage location on the destination device. This first storage location may be a staging area or backup store configured to temporarily store data for further processing.

(119) In operation **716**, the manifest may be transferred from the source device to the destination device. The manifest includes information describing the data objects and relative locations of the data objects on the source device for use in relocating the data objects to corresponding storage locations once stored to the first storage location on the destination device. In one approach, the manifest may include indication of a total number of the data objects that will be transferred from the source device to the destination device.

(120) In operation **718**, the destination device may be rebooted into a configuration mode in response to the input to configure the destination device. The configuration mode allows for exclusive access to a storage medium of the destination device, such as the first storage location. The configuration mode is entered after rebooting the destination device, without user interaction causing the destination device to reboot or enter the configuration mode.

(121) In operation **720**, while in the configuration mode, the destination device may move each of the data objects in the first storage location to second storage locations corresponding to their respective relative locations indicated in the manifest. The movement of the data objects may take place because in configuration mode, the process executing the movement commands has exclusive access to memory of the destination device.

(122) In operation **722**, the destination device may reboot in response to completing moving the data objects to the second storage locations. This allows the destination device to be configured in order to access the data objects in their respective relative locations. This second reboot may also take place without user interaction, such as logging in, providing credentials, etc., thereby making the whole migration process seamless for the user with only the first interaction being needed to start the process.

(123) In an approach, method **700** may include determining whether an OS update is available for the destination device. For example, the OS may have a security patch, feature enhancement, or some other update that has not been performed but is available for the user. In response to determining that the update is available, the destination device may update the first OS of the destination device prior to transferring the data objects.

(124) In one implementation, method **700** may include establishing, by the destination device, a synchronization process between a remote or cloud storage and at least some of the data objects that were previously synchronized with the remote or cloud storage on the source device. In this implementation, after transferring over the data objects that were synchronized with the remote or cloud storage, the automatic backup to the remote or cloud storage may be re-established to ensure that any changes to the synchronized data objects will be captured in the back-up location.

(125) According to an example, method **700** may include determining, by the destination device, a size of the data objects. This size is a measurement of how much storage space is required to store the data objects on the destination device. Further, method **700** may include determining, by the destination device, whether an amount of available storage on the destination device is sufficient to store the data objects to the destination device. In this example, the destination device may transfer the data objects from the source device in response to a determination that the amount of available storage on the destination device is sufficient to store the data objects to the destination device, and the destination device may not transfer any data from the source device in response to a determination that the amount of available storage on the destination device is insufficient to store the data objects.

Graphical User Interfaces

(126) This disclosure above describes various Graphical User Interfaces (GUIs) for implementing

various features, processes or workflows. These GUIs can be presented on a variety of electronic devices including but not limited to laptop computers, desktop computers, computer terminals, television systems, tablet computers, e-book readers and smart phones. One or more of these electronic devices can include a touch-sensitive surface. The touch-sensitive surface can process multiple simultaneous points of input, including processing data related to the pressure, degree or position of each point of input. Such processing can facilitate gestures with multiple fingers, including pinching and swiping.

(127) When the disclosure refers to “select” or “selecting” user interface elements in a GUI, these terms are understood to include clicking or “hovering” with a mouse or other input device over a user interface element, or touching, tapping or gesturing with one or more fingers or stylus on a user interface element. User interface elements can be virtual buttons, menus, selectors, switches, sliders, scrubbers, knobs, thumbnails, links, icons, radio buttons, checkboxes and any other mechanism for receiving input from, or providing feedback to a user.

Privacy

(128) As described above, one aspect of the present technology is the gathering and use of data available from various sources to improve automatic data migration and configuration services. The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, twitter ID's, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other identifying or personal information.

(129) The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, the personal information data can be used to migrate targeted content that is of greater interest to the user. Accordingly, use of such personal information data enables users to calculate control of the migrated data. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.

(130) The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

(131) Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, in the case of automatic data migration and configuration services, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services or anytime thereafter. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

(132) Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

(133) Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data. For example, data may be selected and migrated to the new device by inferring preferences based on non-personal information data or a bare minimum amount of personal information, such as the content being requested by the device associated with a user, other non-personal information available to the automatic data migration and configuration, or publicly available information.

Example System Architecture

(134) FIG. 8 is a block diagram of an example computing device **800** that can implement the features and processes of FIGS. 1A-7. The computing device **800** can include a memory interface **802**, one or more data processors, image processors and/or central processing units **804**, and a peripherals interface **806**. The memory interface **802**, the one or more processors **804** and/or the peripherals interface **806** can be separate components or can be integrated in one or more integrated circuits. The various components in the computing device **800** can be coupled by one or more communication buses or signal lines.

(135) Sensors, devices, and subsystems can be coupled to the peripherals interface **806** to facilitate multiple functionalities. For example, a motion sensor **810**, a light sensor **812**, and a proximity sensor **814** can be coupled to the peripherals interface **806** to facilitate orientation, lighting, and proximity functions. Other sensors **816** can also be connected to the peripherals interface **806**, such as a global navigation satellite system (GNSS) (e.g., GPS receiver), a temperature sensor, a biometric sensor, magnetometer or other sensing device, to facilitate related functionalities.

(136) A camera subsystem **820** and an optical sensor **822**, e.g., a charged coupled device (CCD) or a complementary metal-oxide semiconductor (CMOS) optical sensor, can be utilized to facilitate camera functions, such as recording photographs and video clips. The camera subsystem **820** and the optical sensor **822** can be used to collect images of a user to be used during authentication of a user, e.g., by performing facial recognition analysis.

(137) Communication functions can be facilitated through one or more wireless communication subsystems **824**, which can include radio frequency receivers and transmitters and/or optical (e.g., infrared) receivers and transmitters. The specific design and implementation of the communication

subsystem **824** can depend on the communication network(s) over which the computing device **800** is intended to operate. For example, the computing device **800** can include communication subsystems **824** designed to operate over a GSM network, a GPRS network, an EDGE network, a Wi-Fi or WiMax network, and a Bluetooth™ network. In particular, the wireless communication subsystems **824** can include hosting protocols such that the device **100** can be configured as a base station for other wireless devices.

(138) An audio subsystem **826** can be coupled to a speaker **828** and a microphone **830** to facilitate voice-enabled functions, such as speaker recognition, voice replication, digital recording, and telephony functions. The audio subsystem **826** can be configured to facilitate processing voice commands, voiceprinting and voice authentication, for example.

(139) The I/O subsystem **840** can include a touch-surface controller **842** and/or other input controller(s) **844**. The touch-surface controller **842** can be coupled to a touch surface **846**. The touch surface **846** and touch-surface controller **842** can, for example, detect contact and movement or break thereof using any of a plurality of touch sensitivity technologies, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with the touch surface **846**.

(140) The other input controller(s) **844** can be coupled to other input/control devices **848**, such as one or more buttons, rocker switches, thumb-wheel, infrared port, USB port, and/or a pointer device such as a stylus. The one or more buttons (not shown) can include an up/down button for volume control of the speaker **828** and/or the microphone **830**.

(141) In one implementation, a pressing of the button for a first duration can disengage a lock of the touch surface **846**; and a pressing of the button for a second duration that is longer than the first duration can turn power to the computing device **800** on or off. Pressing the button for a third duration can activate a voice control, or voice command, module that enables the user to speak commands into the microphone **830** to cause the device to execute the spoken command. The user can customize a functionality of one or more of the buttons. The touch surface **846** can, for example, also be used to implement virtual or soft buttons and/or a keyboard.

(142) In some implementations, the computing device **800** can present recorded audio and/or video files, such as MP3, AAC, and MPEG files. In some implementations, the computing device **800** can include the functionality of an MP3 player, such as an iPod™.

(143) The memory interface **802** can be coupled to memory **850**. The memory **850** can include high-speed random-access memory and/or non-volatile memory, such as one or more magnetic disk storage devices, one or more optical storage devices, and/or flash memory (e.g., NAND, NOR). The memory **850** can store an operating system **852**, such as Darwin, RTXC, LINUX, UNIX, OS X, WINDOWS, or an embedded operating system such as Vx Works.

(144) The operating system **852** can include instructions for handling basic system services and for performing hardware dependent tasks. In some implementations, the operating system **852** can be a kernel (e.g., UNIX kernel). In some implementations, the operating system **852** can include instructions for performing voice authentication. For example, operating system **852** can implement the automatic data migration and configuration features as described with reference to FIGS. 1A-7.

(145) The memory **850** can also store communication instructions **854** to facilitate communicating with one or more additional devices, one or more computers and/or one or more servers. The memory **850** can include graphical user interface instructions **856** to facilitate graphic user interface processing; sensor processing instructions **858** to facilitate sensor-related processing and functions; phone instructions **860** to facilitate phone-related processes and functions; electronic messaging instructions **862** to facilitate electronic-messaging related processes and functions; web browsing instructions **864** to facilitate web browsing-related processes and functions; media processing instructions **866** to facilitate media processing-related processes and functions; GNSS/Navigation instructions **868** to facilitate GNSS and navigation-related processes and instructions; and/or

camera instructions **870** to facilitate camera-related processes and functions.

(146) The memory **850** can store software instructions **872** to facilitate other processes and functions, such as the automatic data migration and configuration processes and functions as described with reference to FIGS. **1A-7**.

(147) The memory **850** can also store other software instructions **874**, such as web video instructions to facilitate web video-related processes and functions; and/or web shopping instructions to facilitate web shopping-related processes and functions. In some implementations, the media processing instructions **866** are divided into audio processing instructions and video processing instructions to facilitate audio processing-related processes and functions and video processing-related processes and functions, respectively.

(148) Each of the above identified instructions and applications can correspond to a set of instructions for performing one or more functions described above. These instructions need not be implemented as separate software programs, procedures, or modules. The memory **850** can include additional instructions or fewer instructions. Furthermore, various functions of the computing device **800** can be implemented in hardware and/or in software, including in one or more signal processing and/or application specific integrated circuits.

Claims

1. A method comprising: identifying a set of data objects stored at a source device; selecting a subset of data objects, from the set of data objects for transferring from the source device to a destination device; generating a snapshot of at least one of: the subset of the data objects stored at the source device; or the set of data objects stored at the source device; determining relative locations of data objects, in the subset of the data objects, in relation to other data objects in the subset of the data objects based on the snapshot; generating a manifest indicating the relative locations of data objects, in the subset of the data objects, in relation to other data objects in the subset of the data objects; transferring the manifest to the destination device; and transferring the subset of data objects to the destination device, wherein the method is executed by at least one device including a hardware processor.
2. The method of claim 1, wherein the method is executed by the source device.
3. The method of claim 1, wherein the destination device stores the subset of data objects at respective memory locations at the destination device while maintaining the relative locations of the data objects, in the subset of the data objects, in relation to other data objects in the subset of the data objects.
4. The method of claim 1, wherein the set of data objects comprises data relating to at least one of: a file, settings, a user preference, an image, a photo, a video, a message, account information, a configuration, music, or an application.
5. The method of claim 1, further comprising: requesting, by the source device, prior to transferring the manifest or the subset of data objects, that the destination device update an operating system.
6. A non-transitory computer readable medium comprising instructions which, when executed by one or more hardware processors, causes performance of operations comprising: identifying a set of data objects stored at a source device; selecting a subset of data objects, from the set of data objects for transferring from the source device to a destination device; generating a snapshot of at least one of: the subset of the data objects stored at the source device; or the set of data objects stored at the source device; determining relative locations of data objects, in the subset of the data objects, in relation to other data objects in the subset of the data objects based on the snapshot; generating a manifest indicating the relative locations of data objects, in the subset of the data objects, in relation to other data objects in the subset of the data objects; transferring the manifest to the destination device; and transferring the subset of data objects to the destination device.
7. The non-transitory computer readable medium of claim 6, wherein the operations are performed

by the source device.

8. The non-transitory computer readable medium of claim 6, wherein the destination device stores the subset of data objects at respective memory locations at the destination device while maintaining the relative locations of the data objects, in the subset of the data objects, in relation to other data objects in the subset of the data objects.

9. The non-transitory computer readable medium of claim 6, wherein the set of data objects comprises data relating to at least one of: a file, settings, a user preference, an image, a photo, a video, a message, account information, a configuration, music, or an application.

10. The non-transitory computer readable medium of claim 6, wherein the operations further comprise: requesting, by the source device, prior to transferring the manifest or the subset of data objects, that the destination device update an operating system.

11. A system comprising: at least one device including a hardware processor; the system being configured to perform operations comprising: identifying a set of data objects stored at a source device; selecting a subset of data objects, from the set of data objects for transferring from the source device to a destination device; generating a snapshot of at least one of: the subset of the data objects stored at the source device; or the set of data objects stored at the source device; determining relative locations of data objects, in the subset of the data objects, in relation to other data objects in the subset of the data objects based on the snapshot; generating a manifest indicating the relative locations of data objects, in the subset of the data objects, in relation to other data objects in the subset of the data objects; transferring the manifest to the destination device; and transferring the subset of data objects to the destination device.

12. The system of claim 11, wherein the operations are performed by the source device.

13. The system of claim 11, wherein the destination device stores the subset of data objects at respective memory locations at the destination device while maintaining the relative locations of the data objects, in the subset of the data objects, in relation to other data objects in the subset of the data objects.

14. The system of claim 11, wherein the set of data objects comprises data relating to at least one of: a file, settings, a user preference, an image, a photo, a video, a message, account information, a configuration, music, or an application.
